

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Open-Label, Uncontrolled Study To Evaluate Pharmacokinetics, Efficacy, Safety, Tolerability, And Pharmacodynamics Of Satralizumab In Pediatric Patients With AQP4 Antibody Positive Neuromyelitis Optica Spectrum Disorder (NMOSD) **Short Title/Acronym:** SAKuraSun Study **Protocol Number:** WN41733 **NCT Number:** NCT05199688 **Sponsor Name:** Hoffmann-La Roche (F. Hoffmann-La Roche Ltd / Roche) **Investigational Product:** Satralizumab (Trade Name: Enspryng | ATC Code: L04AC19 | Anti-IL-6 receptor monoclonal antibody) **Phase of Development:** Phase III **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) of satralizumab in pediatric patients aged 2 to 11 years with anti-aquaporin-4 (AQP4) antibody seropositive neuromyelitis optica spectrum disorder (NMOSD). **Secondary Objectives:** To descriptively evaluate the efficacy (e.g., relapse prevention), safety, tolerability, and pharmacodynamics (PD) of satralizumab in this pediatric population.

2.2 Background & Rationale To address the unmet medical need for effective, targeted relapse prevention in pediatric NMOSD patients. While satralizumab has demonstrated a favorable efficacy and safety profile in adults and adolescents with AQP4-IgG seropositive NMOSD, its pharmacokinetic profile, safety, and optimal dosing in children aged 2-11 years require thorough clinical evaluation. This study aims to provide critical data to extend evidence-based treatments to this vulnerable younger population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm, Uncontrolled **Blinding:** Open-label **Randomization:** N/A (Weight-based cohort assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Up to 8 pediatric participants **Number of Sites:** Multicenter (Global sites including Europe, e.g., UK, France, Poland, Italy) **Estimated Study Start:** 2022 **Estimated Primary Completion:** September 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age at screening 2–11 years, inclusive. 2. Body weight at screening ≥ 10 kg. 3. Diagnosed with NMOSD with AQP4 antibody seropositive status (per Wingerchuk 2015 criteria) with clinical evidence of ≥ 1 documented attack (including first attack) in the last year prior to screening. 4. Neurological stability for ≥ 30 days prior to screening and baseline, with an Expanded Disability Status Scale (EDSS) of 0 to 6.5. 5. For patients receiving a baseline immunosuppressant treatment and planning to continue on these therapies, treatment must be at a stable dose for 4 weeks prior to baseline. 6. For female patients of childbearing potential (postmenarcheal): agreement to either remain completely abstinent (refrain from heterosexual intercourse) or to use a reliable means of contraception.

4.2 Exclusion Criteria 1. Pregnancy or lactation. 2. Evidence of other demyelinating diseases mimicking NMOSD. 3. Active or presence of recurrent bacterial, viral, fungal, mycobacterial infection, or other infection at baseline, including evidence of chronic active hepatitis B or C, or untreated latent or active tuberculosis (TB). 4. Receipt of a live or live-attenuated vaccine within 6 weeks prior to baseline. 5. History of severe allergic reaction to a biologic agent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Weight-based dosing: * 10 to <20 kg: 60 mg every 6 weeks. * 20 to <40 kg: 60 mg at Weeks 0, 2, and 4, then every 4 weeks (Q4W). * ≥ 40 kg: 120 mg at Weeks 0, 2, and 4, then every 4 weeks (Q4W).

Route of Administration: Subcutaneous (SC) Injection **Duration of Treatment:** Minimum 48-week main treatment period, followed by an optional satralizumab extension (OSE) period (up to 3.5 years), and a 24-week safety follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of background immunosuppressant treatments (must be stable for ≥ 4 weeks prior to baseline). **Prohibited:** Live or live-attenuated vaccines within 6 weeks before starting treatment and throughout the trial duration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Prior to Week 0 Informed Consent/ Assent, Medical History, AQP4 Status, EDSS Baseline Week 0
First SC Dose,	PK/PD blood draw,	Baseline Safety Assessment	
Interim	Weeks 2, 4, & Q4W/Q6W	Weight-based SC injections, AE	

Check-ups, PK/PD monitoring, Phone check-ups every 2-3 weeks | |
End of Study | Week 48 | End of main treatment phase assessments,
Option to enter OSE | | **Safety Follow-Up** | 24 Weeks Post-Dose |
Clinic visits at 4/6, 12, and 24 weeks post-last dose for
monitoring |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetics (PK) of satralizumab in pediatric patients. **Measurement Method:** Evaluated through the summary of observed serum concentration [C_{through}] of satralizumab, Apparent clearance [CL/F] of satralizumab, and Apparent volume of distribution [V/F] of satralizumab.

7.2 Secondary & Exploratory Endpoints * **Proportion of relapse-free patients** by Week 48. * **Annualized relapse rate (ARR)**, defined as the average number of relapses for each year of the study. * **Time to first relapse (TFR)** after randomization, defined as the time from randomization until the first occurrence of relapse, as determined by the investigator. * **Pharmacodynamic (PD) parameters** (e.g., IL-6 receptor saturation). * **Safety and tolerability** incidence of Adverse Events (AEs) and Serious Adverse Events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via regular clinic visits and phone check-ups roughly every 2–3 weeks.
Serious Adverse Events (SAEs): Standard protocol collection; reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent or internal clinical endpoint committee (CEC) standard utilized for adjudicating NMOSD relapses and reviewing pediatric safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety and PK evaluation sets for patients receiving ≥ 1 dose of satralizumab. **Sample Size Justification:** Small sample size ($N=8$) is standard for ultra-rare pediatric sub-populations. Data is evaluated descriptively without formal comparative power calculations. **Handling of Missing Data:** Standard handling for descriptive summaries of continuous variables and missing safety follow-ups.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-touch monitoring including frequent onsite and remote/phone check-ins to ensure protocol adherence and pediatric safety. **Audit Trail:** eCRF tracking and comprehensive site management overseen by sponsor operations.

11. Study Identifiers & Governance

Primary ID: WN41733 **Registry Number:** NCT05199688 **Posting ID:** 195758279503809597 **Sponsor/Lead Organization:** Hoffmann-La Roche
Study Title: SakuraSun **Official Regulatory Title:** A Phase III, Multicenter, Open-Label, Uncontrolled Study To Evaluate Pharmacokinetics, Efficacy, Safety, Tolerability, And Pharmacodynamics Of Satralizumab In Pediatric Patients With AQP4 Antibody Positive Neuromyelitis Optica Spectrum Disorder (NMOSD)
EMA Documentation: [EPAR Product Information](#)

12. Clinical Framework (NMOSD Specific)

Primary Condition: Neuromyelitis Optica Spectrum Disorder (NMOSD) / Preferred Name: Neuromyelitis Optica **Semantic Type:** Disease or Syndrome **Condition Definition:** An autoimmune inflammatory syndrome characterized by optic neuritis and myelitis. Signs and symptoms include loss of vision, weakness and paralysis of the extremities, and loss of sensation. **Disease Ontology Codes:** MeSH Code: D009471 | SNOMED ID: 25044007
Diagnosis Threshold: AQP4 antibody seropositive status with ≥ 1 documented attack **Medical Methodology:** Interleukin-6 (IL-6) receptor monoclonal antibody inhibition **Optimization Target:** Prevention of autoimmune demyelinating relapses/attacks

13. Study Procedures & Methodology

Management Pathway: Weight-based IL-6 receptor antagonist dosing with or without permitted background immunosuppressive therapy. **Education Protocol (HCP):** Strict adherence to pediatric weight-tiered dosing protocols. **Education Protocol (Patient/Caregiver):** Detailed caregiver informed consent/assent and instructions on safety phone follow-ups and contraception rules for post-menarchal female patients. **Quality Audit Mechanism:** Tracking of protocol-defined relapses and PK sampling compliance.

14. Observation & Follow-Up Schedule

Total Duration: Minimum 48-week primary phase + up to 3.5 years (Optional Extension) + 24 weeks safety follow-up. **Onsite Visit Cadence:** Baseline, Weeks 2, 4, then Q4W or Q6W (based on weight group). **Remote Monitoring Frequency:** Caregiver phone check-ups

every 2-3 weeks. **Checklist Review Interval:** Routine review of ongoing safety events and concomitant medications.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				